

Customer Shopping Behavior Analysis

An end-to-end data project analyzing **3,900 transactions** across product categories to uncover spending patterns, customer segments, and strategic business opportunities.

Dataset at a Glance

3,900

Transactions

Rows in the dataset

18

Features

Columns of data

37

Missing Values

In Review Rating column

\$59.76

Avg. Purchase

Mean purchase amount (USD)

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season), and shopping behavior (Discount Applied, Frequency, Review Rating, Shipping Type).

Data Preparation in Python

→ Load & Explore

Imported dataset via pandas; used `df.info()` and `.describe()` for structure and summary statistics.

→ Clean & Standardize

Imputed missing Review Ratings using median per category; renamed columns to snake_case for consistency.

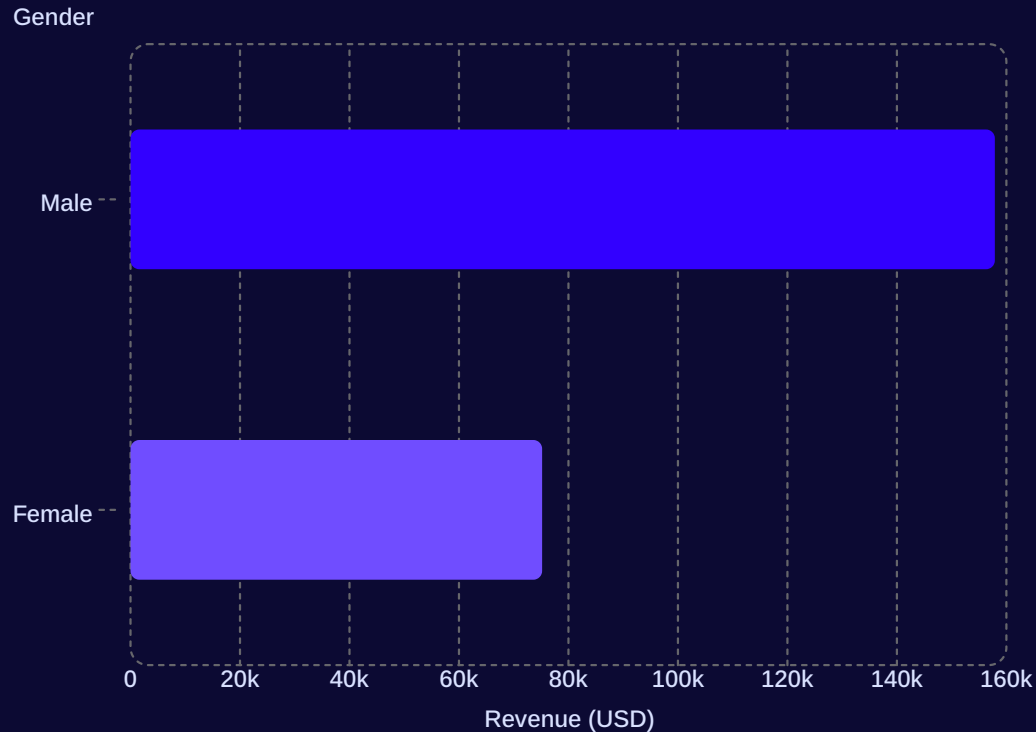
→ Feature Engineering

Created `age_group` by binning ages and `purchase_frequency_days`; dropped redundant `promo_code_used` column.

→ Database Integration

Connected Python to PostgreSQL and loaded the cleaned DataFrame for SQL analysis.

Revenue & Spending Insights



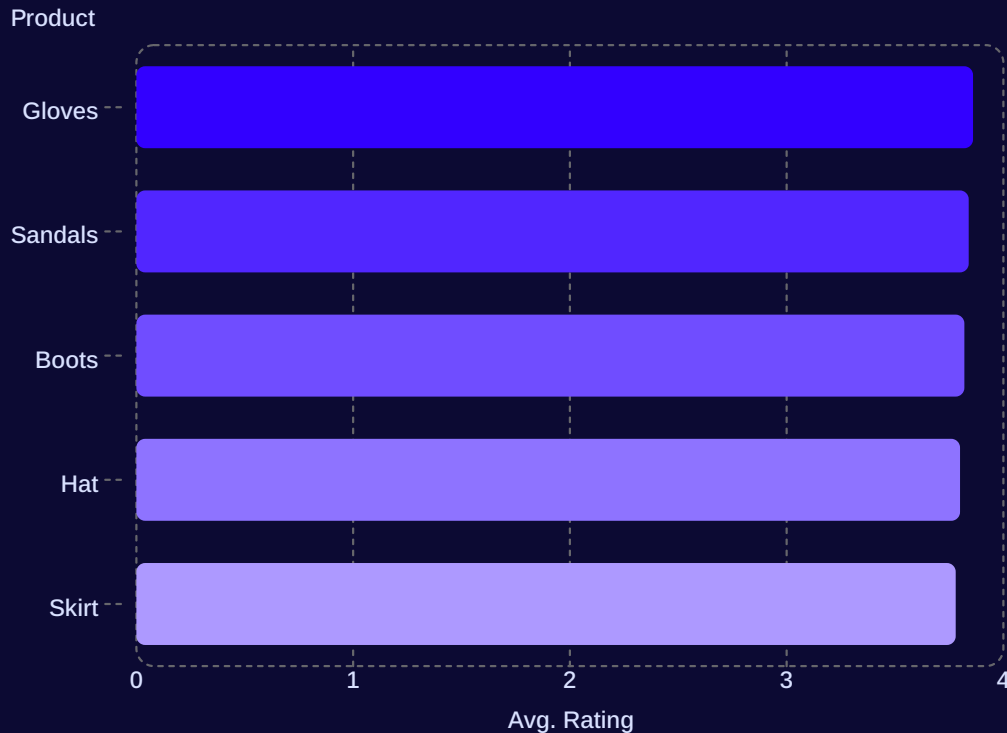
Revenue by Gender

Male customers generated **\$157,890** in revenue — more than double female customers at **\$75,191**.

High-Spending Discount Users

839 customers used discounts yet still spent above the average purchase amount, indicating deal-savvy high-value shoppers.

Products & Shipping Analysis



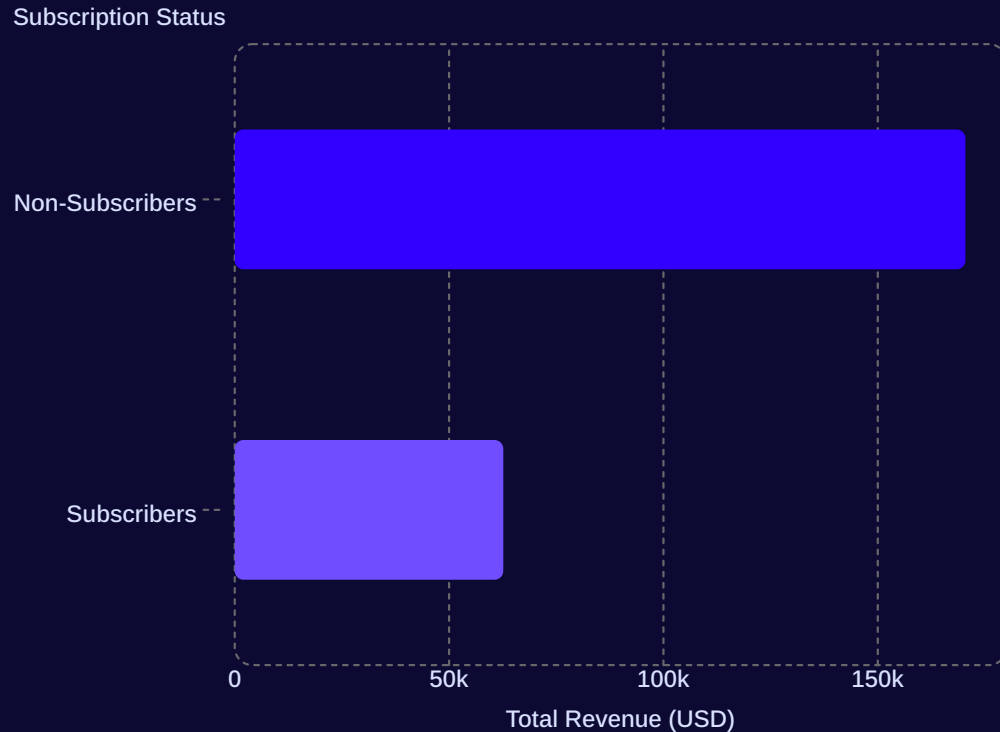
Top 5 Products by Rating

Gloves lead with a **3.86** average rating. All top 5 products cluster tightly between 3.78–3.86.

Shipping Type Comparison

Express shipping customers spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**), suggesting higher intent among express buyers.

Subscription Behavior



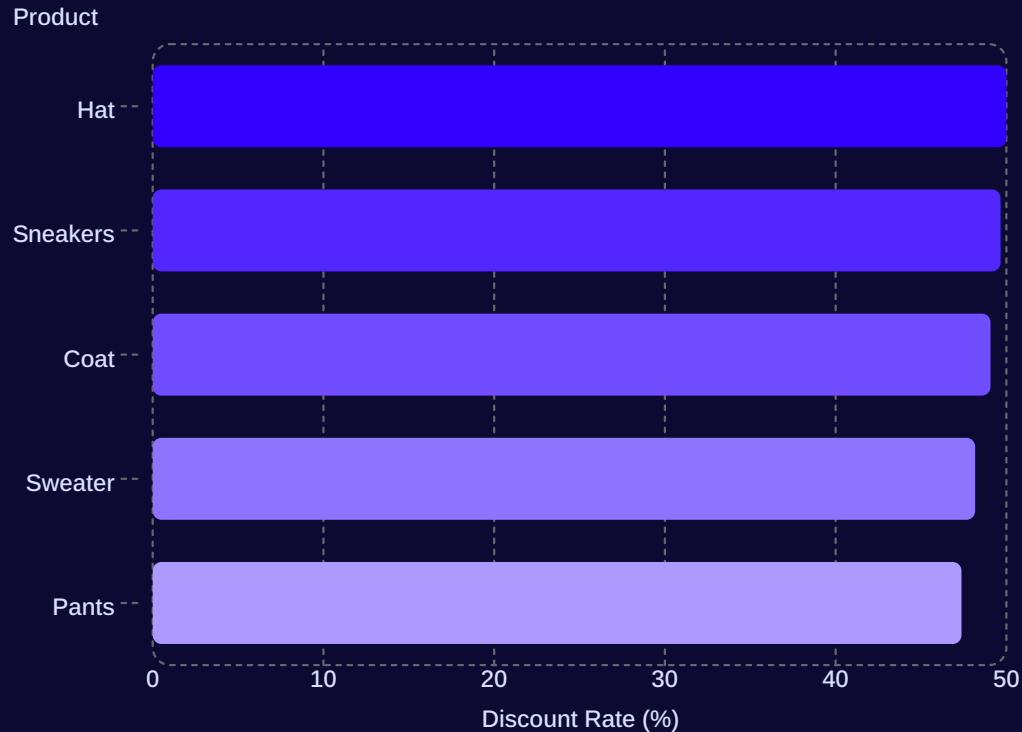
Subscribers vs. Non-Subscribers

Only **27%** of customers are subscribers (1,053), yet they spend nearly the same average amount (**\$59.49** vs. **\$59.87**). Non-subscribers drive **73%** of total revenue.

Repeat Buyers & Subscriptions

Of customers with **>5 purchases**, 958 are subscribers vs. 2,518 non-subscribers — a clear opportunity to convert repeat buyers.

Discount-Dependent Products

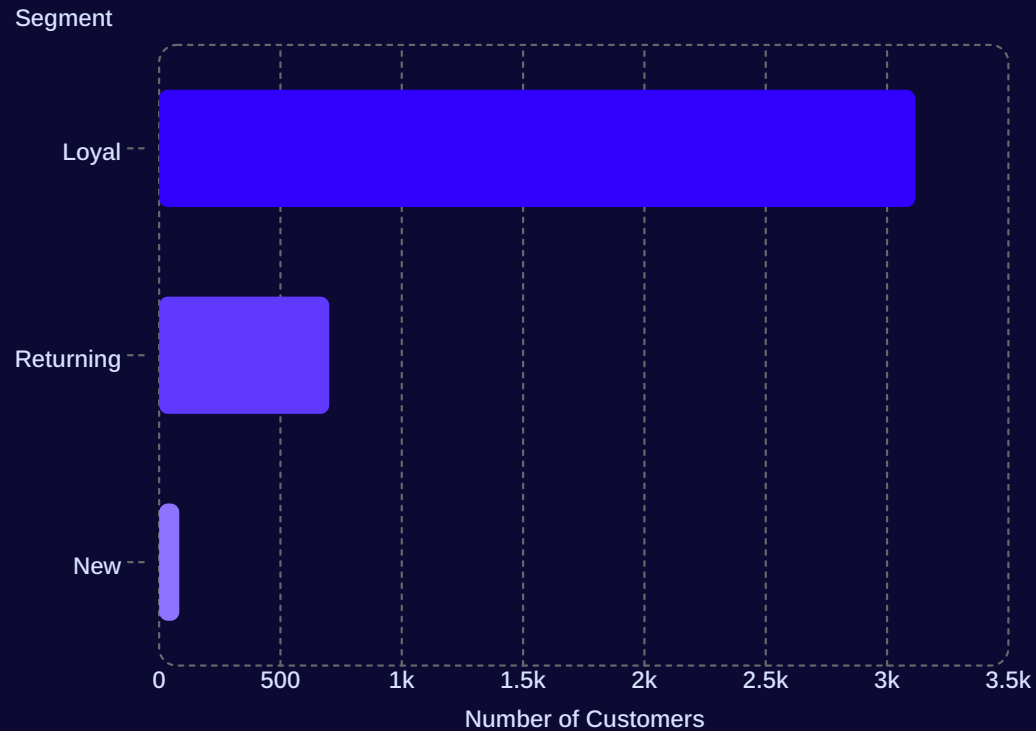


Top 5 Discount-Dependent Products

All five products have ~47–50% of purchases made with a discount applied, suggesting these categories are highly price-sensitive.

- 📌 Hat leads at exactly **50%** discount rate — half of all Hat purchases involve a discount.

Customer Segmentation



Segment Breakdown

Customers were classified into three segments based on purchase history:

Loyal – 3,116

80% of the customer base

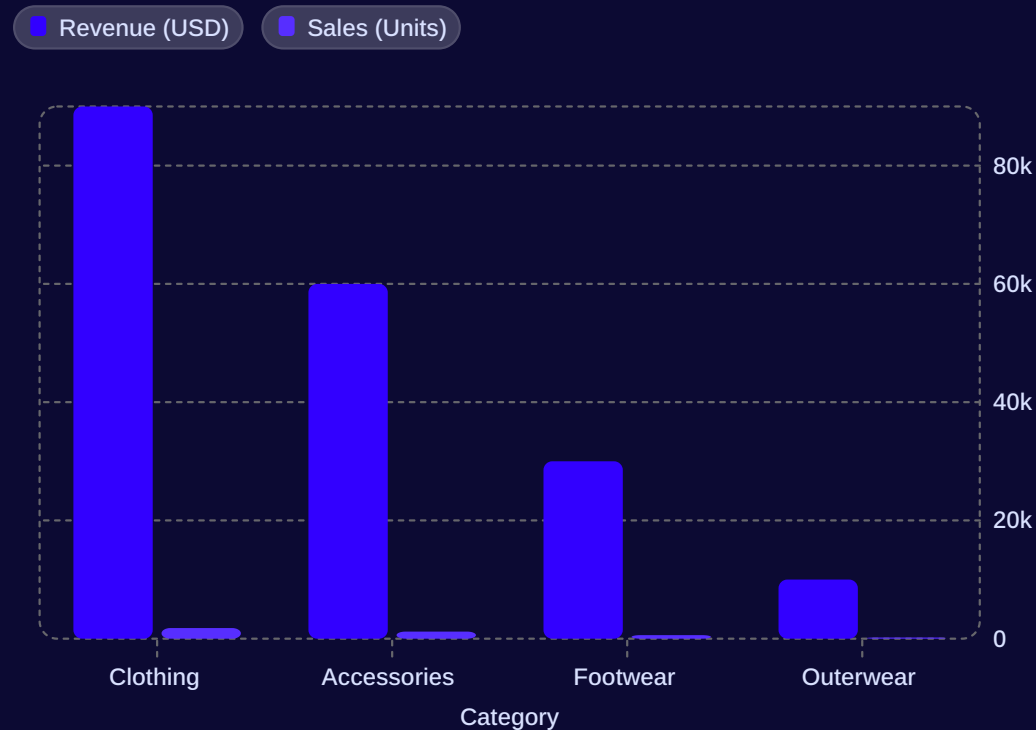
Returning – 701

18% of customers

New – 83

Only 2% are new buyers

Revenue by Category & Age Group



Category Performance

Clothing dominates with ~\$90K revenue and 1,800 units. Top items per category: Blouse (Clothing), Jewelry (Accessories), Sandals (Footwear), Jacket (Outerwear).

Revenue by Age Group

Young Adults lead at **\$62,143**, followed by Middle-aged (**\$59,197**), Adult (**\$55,978**), and Senior (**\$55,763**) — a relatively even spread.

Power BI Dashboard & Recommendations

An interactive **Power BI dashboard** visualizes key metrics: 3.9K customers, \$59.76 avg. purchase, 3.75 avg. rating, with filters for subscription, gender, category, and shipping type.

Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to move them into the "Loyal" segment.

Review Discounts

Balance sales boosts with margin control on discount-heavy products.

Targeted Marketing

Focus on high-revenue age groups and highlight top-rated products in campaigns.

